AIAC lab assignment 18.1

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Q1:Lab Question 1: Weather Forecasting API
A travel company wants to show real-time weather updates for its customers. You are given access to a public weather API that requires an API key and provides weather data in JSON format.
Task 1:
Use AI-assisted coding to write a script that fetches the current temperature and weather description for a given city. The script should handle errors if the API key is invalid or missing.
Task 2:
Extend the script to save the weather data into a local CSV file, ensuring that duplicate entries are avoided. Implement error handling for file I/O exceptions.

Promt:-

Write a Python script that fetches real-time weather data from the OpenWeatherMap API for a given city using an API key. The script should:

1. Display the current temperature and weather description.
2. Handle errors for invalid/missing API keys and city names.
3. Save weather data (city, temperature, description, timestamp) into a local CSV file without duplicates.
4. Implement error handling for network and file I/O exceptions.

```
import requests
import csv
from datetime import datetime
API_KEY = "your_api_key_here"
BASE_URL = "https://api.openweathermap.org/data/2.5/weather"
CSV_FILE = "weather_data.csv"

def fetch_weather(city):
```

```
    try:
        params = {"q": city, "appid": API_KEY, "units": "metric"}
        response = requests.get(BASE_URL, params=params)
        response.raise_for_status()  # Raise an error for bad status codes

        data = response.json()
        if data.get("cod") != 200:
            print(f"Error: {data.get('message', 'Unknown error')}")
            return None

        temperature = data["main"]["temp"]
        description = data["weather"][0]["description"]
        print(f"\n Weather in {city}: {temperature}°C, {description}")

        return {
            "city": city,
            "temperature": temperature,
            "description": description,
            "timestamp": datetime.now().strftime("%Y-%m-%d %H:%M:%S"),
        }

    except requests.exceptions.RequestException as e:
        print(f"Network error: {e}")
        return None

def save_to_csv(weather_data):
    try:
              existing_entries = set()
        try:
            with open(CSV_FILE, mode="r", newline="", encoding="utf-8") as file:
                reader = csv.DictReader(file)
                for row in reader:
                    existing_entries.add((row["city"], row["timestamp"]))
        except FileNotFoundError:
            pass
          if (weather_data["city"], weather_data["timestamp"]) not in existing_entries:
```

```
            with open(CSV_FILE, mode="a", newline="", encoding="utf-8") as file:
                writer = csv.DictWriter(file, fieldnames=["city", "temperature", "description", "times
                if file.tell() == 0:  # Write header if file is new
                    writer.writeheader()
                writer.writerow(weather_data)
            print(" Weather data saved to CSV.")
        else:
            print(" Duplicate entry — not saved.")
    except Exception as e:
        print(f"File I/O error: {e}")
if __name__ == "__main__":
    city_name = input("Enter city name: ").strip()
    if not API_KEY or API_KEY == "your_api_key_here":
        print(" Error: Missing or invalid API key.")
    else:
        weather = fetch_weather(city_name)
        if weather:
            save_to_csv(weather)
```

Input:-
Enter city name: London

output:- Weather in London: 15°C, broken clouds
Weather data saved to CSV.

Q2:-Lab Question 2: Currency Exchange Rate API
A financial startup needs a tool to convert amounts between currencies using an exchange rate API. However, the API occasionally fails due to server downtime.
Task 1:
Write a script (with AI assistance) that takes user input (amount, source currency, target currency) and fetches the latest exchange rate from the API. Include errors in handling invalid currency codes.
Task 2:
Add logic to retry the request up to three times if the API

call fails due to network or server issues and log all failed attempts into a local error log file.

Prompt:
Write a Python script that takes user input (amount, source currency, and target currency) and uses a currency exchange rate API to convert the amount.

- Handle errors for invalid currency codes.
- Retry the request up to 3 times if the API call fails due to network or server issues.
- Log all failed attempts (with timestamp and reason) to a local text file named `error_log.txt`.
- Display the final converted amount if successful.

Code:-

```
import requests
import time
from datetime import datetime
API_URL = "https://api.exchangerate.host/convert"  # Free and public API
LOG_FILE = "error_log.txt"
MAX_RETRIES = 3
def log_error(message):
    with open(LOG_FILE, "a") as f:
        f.write(f"[{datetime.now()}] {message}\n")
def get_exchange_rate(from_currency, to_currency):
    retries = 0
    while retries < MAX_RETRIES:
        try:
            params = {"from": from_currency.upper(), "to": to_currency.upper()}
            response = requests.get(API_URL, params=params, timeout=5)
            response.raise_for_status()

            data = response.js
            if not data.get("info") or "rate" not in data["info"]:
                print(" Invalid currency code or unavailable data.")
```

```
                log_error(f"Invalid currency code: {from_currency}->{to_currency}")
                return None

            return data["info"]["rate"]

        except requests.exceptions.RequestException as e:
            retries += 1
            print(f" Attempt {retries} failed: {e}")
            log_error(f"Attempt {retries} failed: {e}")
            time.sleep(2)  # wait before retry

    print(" Failed to fetch exchange rate after 3 attempts.")
    return None
def convert_currency():
    try:
        amount = float(input("Enter amount: "))
        from_currency = input("Enter source currency (e.g., USD): ").strip().upper()
        to_currency = input("Enter target currency (e.g., INR): ").strip().upper()

        rate = get_exchange_rate(from_currency, to_currency)

        if rate:
            converted = amount * rate
            print(f"\n {amount} {from_currency} = {converted:.2f} {to_currency}")
        else:
            print(" Conversion failed. Please check logs for details.")

    except ValueError:
        print(" Invalid amount. Please enter a numeric value.")
        log_error("Invalid amount entered by user.")
if __name__ == "__main__":
    convert_currency()
```

input:-
Enter amount: 100
Enter source currency (e.g., USD): USD

Enter target currency (e.g., INR): INR

Output:-
100 USD = 8350.40 INR

Q3:-A news aggregator wants to display the latest technology news headlines using a news API. Sometimes, the API responds slowly or returns incomplete data.

Task 1:Use AI-assisted coding to fetch the top 5 technology headlines and print them neatly in the console. Implement error handling for timeout errors by setting a maximum request time.

Task 2:

Clean and preprocess the headlines by removing special characters and converting text to title case. Handle the scenario where the API response contains empty or null values.

promt:-

Write a Python script that fetches the top 5 technology news headlines from a public News API.

- Handle timeout errors by setting a maximum request time.
- Clean and preprocess headlines by removing special characters and converting them to title case.
- Handle cases where API data is empty or contains null values.
- Print the cleaned headlines neatly in the console.

Code:-

```
import requests

import re

API_KEY = "your_newsapi_key_here"

BASE_URL = "https://newsapi.org/v2/top-headlines"

CATEGORY = "technology"

COUNTRY = "us"  TIMEOUT = 5 def clean_headline(text):

    if not text:

        return "No Title Available"

    text = re.sub(r'[^A-Za-z0-9\s]', '', text)

    return text.title().strip()

def fetch_technology_headlines():

    try:

        params = {

            "category": CATEGORY,
```

```
            "country": COUNTRY,
            "apiKey": API_KEY,
            "pageSize": 5        }
        response = requests.get(BASE_URL, params=params,
timeout=TIMEOUT)
        response.raise_for_status()
        data = response.json()
            articles = data.get("articles")
        if not articles:
            print(" No articles found in response.")
            return []

        cleaned_headlines = []
        for article in articles[:5]:
            title = article.get("title")
            cleaned_title = clean_headline(title)
            cleaned_headlines.append(cleaned_title)
       return cleaned_headlines
    except requests.exceptions.Timeout:
        print(" Request timed out. Please try again later.")
        return []
    except requests.exceptions.RequestException as e:
        print(f" Network error: {e}")
        return []
```

```
    except Exception as e:
        print(f" Unexpected error: {e}")
        return []
def main():
    print("Fetching Top 5 Technology Headlines...\n")
    headlines = fetch_technology_headlines()
      if headlines:
        print(" Latest Technology Headlines:\n")
        for i, headline in enumerate(headlines, start=1):
            print(f"{i}. {headline}")
    Else:
      print(" No headlines available to display.")


if __name__ == "__main__":
    main()
```

Input:-

Fetching Top 5 Technology Headlines...

Latest Technology Headlines:

1. Apple Launches New M4 Chip For Macbook Pro
2. Google Introduces Ai Assistant For Developers
3. Microsoft Updates Windows With Enhanced Security
4. Tesla Expands Battery Production In Europe

5. Openai Releases New Chatbot For Businesses